Ying Siu (Sheryl) Liang, Ph.D.

Robotics Software Engineer

sheryl@liang.at
ysl208.github.io
Location: Vienna, Austria

Summary

I specialize in human-centric AI and intuitive robotic systems. I develop generalizable solutions for robots to operate autonomously in real-world environments, making complex robotic systems accessible to non-technical users and ultimately improving their quality of life.
Keywords: end-user robotics, cognitive robotics, social robotics, human-robot interaction

Education

2016 - 2019 **Ph.D. in Robotics**, *Université Grenoble Alpes*, France.
Thesis: *End-User Robot Programming in Cobotic Environments.*

2015 - 2016 **Master of Science (M.S.) in Informatics**, *Grenoble INP - ENSIMAG*, France.
Specialization: Artificial Intelligence and Web.

2008 - 2012 **Integrated Master of Science (MSci) in Mathematics and Computer Science**, *Imperial College London*, United Kingdom, *First Class Honours*.

Experience

2022 - 2026 **Robotics Software Engineer**, *Aeolus Robotics*, Vienna, Austria.
Robot Behavior Planning and Execution (C++, Python)
- Developed solutions for robot reasoning in real-world, user-centric scenarios such as reactive strategies to handle environment changes including failure-handling and recovery behaviors for unpredictable edge-cases.
- Designed performance metrics, Gazebo simulation environments, and rigorous lab testing protocols to ensure robots consistently met and exceeded product requirements.
- Drove cross-functional integration with Manipulation, Vision, and Navigation teams and ensured resilient robot behaviors with live troubleshooting.

2020 - 2021 **Research Scientist**, *A*STAR Institute for High Performance Computing*, Singapore.
CollabAI project in Social and Cognitive Computing (Python, Clojure)
- Developed a novel object permanence model for robot belief management that tracks invisible displacements in dynamic environments and outperformed state-of-the-art models in spatial reasoning and situational awareness. [1,2]
- Led the Cognitive Systems Group in developing and integrating the ICARUS architecture. Integrated high-level reasoning components to combine NLU, perception, and manipulation modules and to drive goal-oriented behaviors on Universal Robots (UR5, UR16e). [3]

Jun-Sep 2019 **Research Intern**, *A*STAR Institute for Infocomm Research*, Singapore.
Inferring the Geometric Nullspace for Robot Skills from Human Demonstrations (C++) [4]

2016 - 2019 **Ph.D. in Robotics**, *Grenoble Computer Science Lab*, Univ. Grenoble Alpes, France.
End-User Robot Programming in Cobotic Environments (C++, JavaScript/Polymer) [5-9]
- GitHub: https://github.com/ysl208/iRoPro
- Pioneered iRoPro, an interactive robot programming framework that lets end-users teach robots new actions by demonstration, enabling them to tackle previously unseen, complex tasks on the fly. Engineered a full-stack robotics system by fusing advanced object perception (point cloud segmentation), manipulation, motion planning (MoveIt), behavior planning (PDDL), and human-computer interaction into a seamless, end-to-end solution.
- Designed and executed human-robot interaction studies to rigorously evaluate user acceptance, usability, and overall system performance.

Jul-Dec 2017 **Research Intern**, *Human-Centered Robotics Lab*, Univ. of Washington, USA.
End-User Programming of Goals and Actions for Robotic Shelf Organisation (C++) [8]

Feb-Jun 2016 **Research Intern**, *Grenoble Computer Science Lab*, Univ. Grenoble Alpes, France.
Robot Programming by Demonstration for Non-Experts - Proof of Concept (Python) [10]

2012 - 2015 **Software Engineer**, *Deutsche Bank Global Technology*, London, United Kingdom.
- Implemented features for data access from external sources, allowing users to query and update information directly within an OLAP Cube through a web application (C#, HTML/JavaScript).

Skills

Programming C++, Python, Clojure, C#, SQL, PHP, HTML/JavaScript

Robotic ROS, RVIZ, Gazebo, OpenCV, PDDL

Frameworks Aeolus Robot, Universal Robot, Baxter Robot, Fetch Robot, QTRobot

Ubuntu Linux (CLI), Jenkins (CI/CD), Docker, Git, Jira, Confluence, VS Code

Spoken German (native), English (bilingual), French (C2), Chinese (B2), Spanish (A1)

Attended Courses

2021 IS5452 Affective Computing, National University of Singapore, Singapore

2018 ROS Summer School, FH Aachen, Germany

Additional Responsibilities

2021 **Team lead**, *Project on Engagement Detection in Virtual Classes*, Singapore.

2021 **Co-coordinator**, *NIE-A*STAR Workshop for AI in Education*, Singapore.

2018 **Mentor of undergraduate interns**, *UI design for iRoPro*, Grenoble, France.

2017 **Teaching assistant**, *Introduction to Databases*, Grenoble, France.

2017 - 2023 **Reviewer**, *HRI'23, IROS'21, ICRA'18,'19,'21, ACII'21, RSS'19, RO-MAN'19, HRI'18*.

Volunteering

2021 **Elementary math tutor**, *South Central Community Service Centre*, Singapore.

2020 - 2021 **Trishaw pilot for nursing homes**, *Cycling Without Age*, Singapore.

2013 - 2015 **GCSE math tutor**, *Volunteering Matters*, London, United Kingdom.

Awards and Honors

2016 - 2019 French Ministry Ph.D. research grant by MSTII Doctoral School (ED MSTII)

Mar 2018 Prize for best presentation at the LIG Ph.D. day

Jul 2017 IDEX International Mobility Grant to collaborate with the Univ. of Washington

Jun 2017 Prize for best presentation at the ED MSTII Ph.D. day

Hobbies

Basketball, Boardgames, Cycling, Table tennis, Snowboarding

Publications and Other Communication

[1] Y.S. Liang, C. Zhang, D. Choi, K. Kwok (2021). **Improving Object Permanence using Action Annotations and Reasoning.** Intl. Conf. on Intelligent Robots and Systems (IROS). Virtual (Oral): IEEE/RSJ Press.

[2] Y.S. Liang, D. Choi, K. Kwok (2021). **Maintaining a Reliable World Model using Action-aware Perceptual Anchoring.** Intl. Conf. on Robotics and Automation (ICRA). Xi'An, China (Oral): IEEE Press.

[3] D. Choi, W. Shi, Y.S. Liang, K.H. Yeo, J.J. Kim (2021). **Controlling Industrial Robots with High-level Verbal Commands.** Intl. Conf. on Social Robotics (ICSR). Singapore: IEEE/RSJ Press.

[4] C. Cai, Y.S. Liang, N. Somani, Y. Wu (2020). **Inferring the Geometric Nullspace of Robot Skills from Human Demonstrations.** Intl. Conf. on Robotics and Automation (ICRA). Virtual: IEEE Press.

[5] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2021). **iRoPro, an interactive Robot Programming Framework.** Intl. Journal of Social Robotics (IJSR). Springer-Verlag.

[6] Y.S. Liang (2019). **End-User Robot Programming in Cobotic Enviroments.** PhD thesis. Grenoble, France.

[7] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2019). **End-User Programming of Low- and High-Level Actions for Robotic Task Planning.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). New Delhi, India (Oral): IEEE Press.

[8] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty, M. Cakmak (2018). **Simultaneous End-User Programming of Goals and Actions for Robotic Shelf Organization.** Intl. Conf. on Intelligent Robots and Systems (IROS). Madrid, Spain (Oral): IEEE/RSJ Press.

[9] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **Evaluation of a Robot Programming Framework for Non-Experts using Symbolic Planning Representations.** Intl. Symposium on Robot and Human Interactive Communication (RO-MAN). Lisbon, Portugal (Poster): IEEE Press.

[10] Y.S. Liang, D. Pellier, H. Fiorino, S. Pesty (2017). **A Framework for Robot Programming in Cobotic Environments: First User Experiments.** Intl. Conf. on Mechatronics and Robotics Engineering (ICMRE). Paris, France (Oral): ACM.

May 2018 **Spotlight talk**, *A Robot Programming Framework for Non-Experts*, Robots in Human Environments (RHUM) Workshop, Grenoble, France.

Jun 2017 **Oral presentation**, *Robot Programming by Demonstration*, Affects, Artificial Companions and Interactions (ACAI) Working Group, Paris, France.